وزارة الاتصالات وتكنولوجيا المعلومات

رواد مصر الرقمية

Digital Egyptian Pioneers innitiative - DEPI

AI and Data Science - Machine learning track

BankWise

Documentation

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Project Proposal

1- Introduction

Background & Context

Retail banking institutions in Egypt handle a high volume of daily customer inquiries regarding personal loans, administrative fees, account policies, and product details. Managing these routine inquiries through traditional channels requires significant operational resources and often creates friction for customers seeking quick answers.

Problem Statement

- Branch Overcrowding: Customers frequently visit physical bank branches and wait in long queues simply to ask basic questions about loan eligibility, fees, or required documents.
- Call-Center Bottlenecks: Customer support hotlines experience fluctuating call volumes and long hold times, while support availability remains restricted to working hours.
- Repetitive Support Workload: Support agents spend repetitive effort addressing standard questions.

Proposed Solution Overview

BankWise AI is a virtual 24/7 banking assistant designed specifically for Egyptian retail banking customers in Egypt. The system combines a deterministic rule-based loan eligibility calculator with a Retrieval-Augmented Generation (RAG) knowledge engine:

- **Exact Financial Logic:** Evaluates loan eligibility and calculates estimated monthly installments using explicit underwriting rules rather than generative model approximation.
- **Grounded Document Search:** Answers product and fee questions strictly using retrieved content from bank FAQs

2- Project Overviewl

BankWise AI provides an automated, reliable, and safe-by-design conversational interface for retail banking inquiries. The project bridges the gap between natural language understanding and strict financial rule compliance, delivering immediate guidance to users without putting operational load on bank staff.

Project Objectives

- Automate 24/7 Customer Inquiries: Deliver instant, accurate answers to routine banking questions across all hours of the day.
- Ensure Deterministic Loan Calculations: Implement exact financial calculations for loan eligibility and monthly installment estimates using external rule evaluation rather than LLM math estimations.
- Eliminate Information Hallucinations: Restrict general policy, product, and fee responses exclusively to verified passages retrieved from official bank policy documents.

2- Project Proposal

1. Arabic Chat Engine

- **Dialectal Processing:** Receives customer inputs in Arabic and returns responses in the same language.
- **Session Context Management:** Maintains conversation context within an active session so users can ask follow-up questions without repeating details.

2. Loan Eligibility & Calculation Engine

- **Parameter Extraction:** Extracts key financial variables from user messages: monthly income, existing monthly debt payments, requested loan amount, tenor (loan duration), and age.
- **Interactive Prompting:** Identifies missing or ambiguous parameters and interactively asks the user for required details before running checks.
- **Input Validation:** Enforces checks on numeric inputs to reject invalid or negative values with clear explanatory feedback.
- **Rule-Based Evaluation:** Evaluates user data against published bank underwriting rules configured in a central file (Debt Burden Ratio (DBR), minimum monthly income, maximum tenor, age limits).
- **Installment Calculation:** Performs exact mathematical calculation of estimated monthly installments.
- **Transparent Decision Explanation:** Displays eligibility decision (Eligible / Non-Eligible) along with specific rule-based reasons and estimated installment amounts.
- **Deterministic Isolation:** Isolates financial calculations completely from LLM generation to eliminate numerical hallucinations.

3. RAG-Powered FAQ & Policy Engine

- **Knowledge Base Ingestion:** Indexing bank policy documents, product guides, and fee schedules using Arabic-capable embeddings into a vector database.
- **Context Retrieval:** Searches and retrieves relevant document passages based on vector similarity matching for user inquiries.
- **Grounded Generation:** Restricts LLM responses strictly to retrieved source text to guarantee accuracy.
- **Source Attribution:** Links generated answers to source document names and retrieval dates.
- **Fallback Handling:** Triggers an explicit refusal statement ("No confirmed information available") and directs users to bank contact channels when retrieved text relevance is insufficient.

4. Administrative Performance

- **Usage Analytics:** Tracks and displays most-frequently asked customer questions.
- **Performance Tracking:** Logs query latency, API token usage, and computed cost per conversation session.

5. Project Plan

The BankWise AI project execution spans five months, from October 1 to December 4. The following subsections summarize the key project milestones, tangibles/deliverables, and strategic resource allocations required to build, evaluate, and deploy the platform.

1. Milestones (Oct 1 – Dec 4)
The project is structured into five core phases aligned with key technical checkpoints:

Phase & Target Date	Milestone	Key Objectives & Focus
Phase 1 (October 31)	**M1: Knowledge Base & Vector Indexing**	Setup Docker, PostgreSQL/pgvector; parse bank policy PDFs and generate Arabic embeddings (GATE-AraBERT/BGE-M3).
Phase 2 (November 30)	**M2: Core Engines & Arabic Normalization**	Develop deterministic loan calculator (CBE rules), RAG query pipeline, and Arabic text pre-processing.
Phase 3 (December 31)	**M3: System Integration & Dashboard**	Implement Redis session memory, finalize FastAPI routing, and complete React client chat & staff admin UI.
Phase 4 (January 31)	**M4: RAGAS Benchmark & Security Audit**	Evaluate RAG accuracy (RAGAS/SILMA benchmarks), verify exact loan logic, and red-team prompt safety/PII masking.
Phase 5 (February 28)	**M5: Production Scaling & DEPI Handover**	Deploy scaled multi-worker Docker setup, verify live latency/cost metrics, and finalize capstone documentation.

2. Key Deliverables

Project outcomes are categorized into Technical, Documentation, and Evaluation deliverables:

Category	Deliverable Name	Summary & Requirements
Technical	**Engineered Software System**	Bilingual Arabic chat engine with Redis session memory, deterministic Python loan calculator (DBR/CBE limits), RAG vector engine, FastAPI backend, and React UI/Admin dashboard.
Documentation	**System & Compliance Manuals**	System Architecture Blueprint, Credit Policy Configuration Manual, Data Privacy & Security Protocol, and complete DEPI Capstone Report.
Evaluation	**Benchmark & Audit Reports**	RAGAS accuracy evaluation report (≥80% accuracy target), loan logic test logs (100% disclaimer compliance), and system latency/security audit.

3. Resource Allocation

3.1 Team Roles & Responsibilities

Role	Primary Responsibilities	Core Allocation
Project Manager / Lead	Schedule tracking, risk mitigation, KPI oversight, and final report consolidation.	Phases 1, 4, 5
AI / NLP Engineer	RAG pipeline architecture, Arabic normalization, embedding selection, and prompt engineering.	Phases 1, 2, 3
Backend Engineer	FastAPI web services, PostgreSQL/pgvector database setup, and loan calculator integration.	Phases 1, 2, 5
Frontend Developer	React-based chat interface and administrative telemetry/cost monitoring dashboard.	Phases 3, 4
QA & Compliance Analyst	CBE policy verification, synthetic Arabic test suite creation, RAGAS benchmarking, and security testing.	Phases 2, 4

3.2 Technical & Data Infrastructure

- **Backend & Database:** FastAPI hosted in Docker containers; PostgreSQL extended with pgvector for high-performance vector search.
- **Session & Memory:** Redis in-memory caching store for ephemeral multi-turn session tracking.
- **AI Models:** GATE-AraBERT-v1 / BGE-M3 for Arabic embeddings; OpenAI GPT-4o / DeepSeek (via REST API) for grounded text generation.
- **Knowledge Resources:** Official Egyptian retail banking policy PDFs, CBE underwriting circulars, and custom-curated Arabic test suites.

Gantt Chart

To access gantt chart: press here

Task Assignment & Roles

Task	Individual(s)	Project Phase, Task Area
Data Collection,Project setup, Use cases, System Design, Presentation.	All Members	Planning, Analysis & Design, Final Reports, and Data Collection
RAG Pipeline	Tarek Mahmoud & Mahmoud Al-Adgham	Implementation: Data Collection & RAG Pipeline
LLM	Rashy Maged	Implementation: LLM Integration & Agent Logic
Deployment & UI	Hady Omar & Ghada Saeed	Implementation: UI Development, Integration, and version Control

Risk Assessment

ID	Risk	Likelihood	Impact	Rating	Mitigation
R-01	Outdated fee or loan information	M	H	High	Record source URL and date per document; show source; refresh on schedule
R-02	LLM invents a figure, rate or decision	M	H	High	Answer only from retrieved text; refuse on low relevance; test with trap questions
R-03	Misleading promise of approval	L	H	Medium	"Preliminary estimate" statement in every eligibility reply; automated phrase check
R-04	Eligibility rules differ from policy, or wrong installment	M	H	High	Rules in one config file reviewed by credit risk; unit tests with boundary values
R-05	Weak Egyptian Arabic understanding and retrieval	H	M	High	Normalize Arabic text; Arabic-capable embeddings; dialect test set
R-06	Loan inputs misread (Arabic digits, "ألف", tenor)	H	M	High	Normalize numbers; validate ranges; confirm extracted values with the customer
R-07	Prompt injection or jailbreak	M	H	High	Separate instructions from user text; treat retrieved text as data; red-team tests
R-08	Sensitive data requested or stored (ID, card)	M	H	High	Never ask for them; mask ID/card patterns; session kept in memory and cleared on reset
R-09	LLM, tool or retrieval failure causes a crash	M	M	Medium	Timeouts, retries and an Arabic fallback message; failure-injection tests
R-10	Cost overrun or abuse	M	M	Medium	Rate limits and token caps; cost per conversation on the dashboard

KPIs

ID	Category	KPI	Target
K-01	Accuracy	Test questions answered correctly	At least 8 of 10
K-02	Accuracy	Invented figures in answers	0
K-03	Accuracy	Eligibility results matching the rules	100% of test cases
K-04	Reliability	Unhandled exceptions in scripted tests	0
K-05	Error handling	Failures showing a clear fallback message	100%
K-06	Privacy	Sensitive data requested or stored	0
K-07	Transparency	Documents with source URL and retrieval date	100%
K-08	Compliance	Replies promising approval; eligibility replies with the estimate statement	0; 100%
K-09	Performance	Average response time	5 s or less
K-10	Cost	Token cost per conversation	Set with sponsor
K-11	Experience	Conversations resolved without a human	70% or more
K-12	Monitoring	Dashboard shows top questions, response time and cost from real data	All shown

Literature Review

To access literature review: press here

Stakeholder Analysis

Stakeholder	Category	Key needs	Main concerns / risks
Customers and loan applicants	Primary	Fast, correct answers at any	Wrong or outdated fee
Support team leads	Primary	Dashboard with most-asked	Metrics that cannot be
Credit risk and product teams	Secondary	Eligibility rules that follow	The model inventing a
Compliance and legal	Secondary	Required disclaimers;	Unapproved advice; regulatory
Information security and IT	Secondary	No hardcoded secrets; rate	Prompt injection; abuse and cost
Retail banking management	Secondary	Lower branch and call-center load;	Reputational damage from a
Project team and evaluators	Secondary	Representative Arabic data and	Poor Arabic retrieval quality;

User Stories

- **Priority uses MoSCoW: M = Must, S = Should, C = Could. Each story lists key acceptance criteria (AC).**

ID	User story	Priority	Acceptance criteria
US-01	As a customer, I want to ask about bank products, fees and policies in Arabic or English so that I get an answer without calling or visiting.	M	Replies in the language of the question; answer is grounded in the knowledge base and replies within a defined latency target

ID	User story	Priority	Acceptance criteria
US-02	As a customer, I want fees explained with concrete amounts so that I understand what I will pay.	M	Fee answers come only from the recorded fee documents; if not found, the bot says so
US-03	As a customer, I want to ask follow-up questions without repeating myself.	M	Session memory keeps the topic and entities for the whole session
US-04	As a customer, I want the bot to say when it does not know so that I am not misled.	M	Low-relevance retrieval triggers a refusal and a suggestion to contact the bank; no invented figures
US-05	As a loan applicant, I want to check my eligibility by giving a few details so that I know whether it is worth applying.	M	Tool applies the configured rules; result shows eligibility, reasons and estimated installment; clear "preliminary estimate" statement
US-06	As an applicant, I want to know why I am not eligible and what I can change so that I can improve my chances.	S	Reasons are rule-based and explainable (for example installment-to-income ratio above the limit)

ID	User story	Priority	Acceptance criteria
US-07	As a risk officer, I want the assistant never to promise approval so that the bank is not committed by a chatbot.	M	No response states or implies a guaranteed approval
US-08	As a team lead, I want to see the most-asked questions so that we can improve content.	M	Ranked list of frequent questions on the dashboard
US-09	As a sponsor, I want token usage and cost per conversation so that I can control the budget.	M	Per-conversation and aggregate token and cost figures from real API usage

Use Cases

UC-01 — Ask an FAQ, product or fee question

- **Primary actor: Customer**
- **Goal: Get an accurate answer from approved bank content**
- **Related stories: US-01 to US-04**

Main flow:

1. **Customer asks a question in Arabic.**
2. **System checks the session memory for context.**
3. **System retrieves the most relevant chunks from the FAQ, policy and fee documents.**
4. **System generates an answer grounded in the retrieved text.**
5. **System shows the answer with its source.**
6. **Customer may ask a follow-up; the flow repeats using session memory.**

Alternate flows:
•No relevant source found: the system says it cannot confirm and suggests contacting the bank.
•Out-of-scope or non-banking question: polite refusal and redirect.
•Question in another language: the system states that the service is currently Arabic-only.

UC-02 — Check loan eligibility
•Primary actor: Loan applicant
•Goal: Receive an indicative eligibility result with the estimated installment
•Related stories: US-05 to US-07
Main flow:
1.Applicant says they want to know if they qualify for a personal loan.
2.System asks for missing inputs one at a time (monthly income, existing debts, loan amount, tenor, age).
3.System validates the inputs and calls the eligibility tool.
4.Tool returns the decision, the reasons and the estimated installment.
5.System explains the result in plain Arabic with a "preliminary estimate, not a final approval" statement.
Alternate flows:
•Not eligible: the system explains the rule-based reasons and what could change the outcome.
•Invalid or inconsistent input: the system asks again with an example.
•Tool unavailable: fallback message with an option to retry later or contact the bank.

.UC-03 — Monitor performance and cost
•Primary actor: Team lead or sponsor
•Goal: Understand usage, speed and cost
•Related stories: US-08, US-09
Main flow:
1.Staff member opens the monitoring dashboard.
2.System shows the most-asked questions, average response time, and token usage and cost per conversation.
3.Staff member reviews the figures to improve content or control budget.
Alternate flows:
•No data yet: the dashboard shows an empty-state message.

Functional and non functional requirements

1- Chat : FR

Recuirements
The system shall provide a chat interface where the customer types a question in Egyptian Arabic and receives a response.
The system shall always respond in Arabic. If the customer writes in another language, it shall politely state that the service is currently Arabic-only.
The system shall keep conversation context within a session, so follow-up questions can refer to earlier messages.
The system shall allow the customer to start a new conversation and clear the current session.

2- Loan Eligibility : FR

Recuirements
The system shall extract the required inputs from the customer’s message: monthly income, existing monthly debts, loan amount, tenor, and age
If any input is missing or unclear, the system shall ask the customer for it before running the check.
The system shall validate inputs (numeric, positive, within sensible ranges) and reject invalid values with a clear message.
The system shall evaluate eligibility using a rule-based tool that applies rules loaded from a configuration file (debt burden ratio, minimum income, maximum tenor, age limits).
The system shall calculate the estimated monthly installment.
The system shall return the decision (eligible or not), the specific reasons, and the estimated installment.

3- FAQ : FR

Recuirements
The system shall answer questions about loans, fees, and products using retrieved passages from the knowledge base of bank documents.
The system shall base answers only on retrieved content.
If no sufficiently relevant content is found, the system shall say it has no confirmed information and suggest contacting the bank.
The system shall allow the customer to start a new conversation and clear the current session.

Non Functional Req.

Category	Recuirements
Accuracy	The system shall answer at least 8 of 10 representative test questions correctly.
Reliability	The system shall produce zero unhandled exceptions across the scripted test conversations, including empty, very long, or unusual input.
Error handling	When the LLM, a tool, or retrieval fails, the user shall receive a clear fallback message instead of a crash or technical error.
Privacy	The system shall not ask for or store sensitive personal data (national ID, card numbers), and session data shall not persist after the session ends.
Transparency	Each knowledge-base document shall have its source URL and retrieval date recorded.